

SOCHI, Russian Federation

WMO: 371710

Lat: 43.450N

Lon: 39.957E

Elev: 27

StdP: 101.00

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	-0.2	-6.8	2.1	1.1	-5.0	2.5	2.6	9.9	8.9	8.4	8.2	3.2	70	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	30.9	23.8	29.8	23.7	28.8	23.2	25.6	29.0	24.8	28.3	24.0	27.6	3.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	19.3	28.1	23.8	18.7	27.6	22.9	17.7	26.8	78.8	29.1	75.6	28.3	72.5	27.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.6	5.9	-4.3	33.7	2.0	2.0	-5.7	35.1	-6.8	36.3	-7.9	37.4	-9.3	38.8
			-5.1	26.7	1.8	0.8	-6.4	27.3	-7.5	27.7	-8.5	28.2	-9.9	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	6.5	6.7	8.7	12.4	16.8	21.0	23.7	24.5	20.7	16.3	11.5	8.0
	DBStd	7.10	3.23	3.52	3.67	3.56	3.28	2.68	2.21	2.13	2.70	3.11	3.30	3.44
	HDD10.0	406	116	103	71	14	0	0	0	0	0	1	22	79
	HDD18.3	1861	367	326	300	181	69	7	0	0	10	76	204	320
	CDD10.0	2154	7	11	29	87	211	329	426	449	322	197	68	18
	CDD18.3	567	0	0	1	3	22	87	168	191	82	14	1	0
	CDH23.3	4027	0	0	4	20	112	460	1323	1608	442	58	1	0
	CDH26.7	897	0	0	1	2	21	72	306	428	63	5	0	0
Wind	WSAvg	2.9	3.3	3.2	3.0	2.7	2.5	2.8	2.8	2.9	2.9	2.8	3.0	3.3
Precipitation	PrecAvg	1582	163	136	136	107	111	99	105	93	142	178	159	154
	PrecMax	2074	253	257	300	192	314	243	215	240	467	446	395	323
	PrecMin	1207	41	32	52	37	21	19	34	5	19	36	15	50
	PrecStd	218	55	56	51	41	69	51	55	59	89	100	93	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	18.5	22.7	25.8	29.0	30.2	32.9	32.2	30.2	27.0	22.2	19.2
		MCWB	9.9	10.3	12.5	15.0	18.6	21.8	23.0	23.6	21.6	18.6	13.7	10.7
	2%	DB	14.9	16.0	19.1	23.0	26.2	28.5	30.9	31.0	28.5	25.0	20.1	16.8
		MCWB	8.8	9.2	11.1	14.3	18.2	22.2	24.2	24.4	21.6	18.6	13.3	10.6
	5%	DB	12.9	14.1	16.9	20.3	24.1	27.1	29.2	30.0	27.1	23.2	18.2	14.8
		MCWB	8.2	8.4	10.4	13.5	17.8	22.0	23.9	24.3	21.5	17.9	13.1	10.2
	10%	DB	11.2	12.2	14.2	18.2	22.2	25.9	28.1	29.0	26.0	21.8	16.8	13.0
		MCWB	7.9	8.1	9.6	12.8	17.7	21.5	23.5	23.9	20.9	17.4	12.7	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	11.5	13.4	16.4	20.8	24.5	26.4	26.6	24.5	20.9	15.7	13.2
		MCDB	14.1	16.1	19.5	22.1	24.5	27.9	29.7	30.1	28.3	24.9	18.9	15.8
	2%	WB	9.9	10.2	12.0	15.2	19.7	23.5	25.5	25.8	22.9	19.4	14.8	11.7
		MCDB	12.6	13.6	17.1	20.2	23.5	26.7	28.9	29.2	26.5	23.2	17.9	14.6
	5%	WB	9.1	9.3	11.0	14.3	18.8	22.6	24.7	25.1	22.0	18.5	13.9	10.7
		MCDB	11.8	12.5	15.1	19.0	22.5	26.0	28.1	28.7	25.9	22.1	17.1	13.7
	10%	WB	8.3	8.5	10.2	13.4	18.1	21.9	23.9	24.4	21.2	17.6	13.0	9.9
		MCDB	10.8	11.6	13.3	17.4	21.7	25.2	27.5	27.9	25.1	20.9	16.2	12.6
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.2	7.2	8.0	7.7	7.5	7.5	7.7	8.3	8.2	8.4	7.3
		MCDBR	8.9	9.9	11.3	11.9	10.7	8.6	8.4	8.2	9.2	10.0	10.4	9.4
		MCWBR	5.2	5.2	5.5	5.7	5.4	5.1	5.0	5.0	5.6	6.1	6.3	5.6
	5% WB	MCDBR	7.5	8.1	9.3	9.9	8.6	7.5	7.6	7.5	8.3	8.8	9.1	8.0
		MCWBR	4.8	4.8	5.0	5.2	4.9	4.8	5.0	5.0	5.4	5.8	6.1	5.2
Clear-Sky Solar Irradiance	taub	0.354	0.388	0.440	0.485	0.453	0.445	0.463	0.466	0.418	0.402	0.362	0.347	
	taud	2.279	2.169	2.055	1.995	2.193	2.277	2.247	2.232	2.343	2.326	2.366	2.327	
	Ebn at Noon	750	783	786	783	825	832	811	794	807	765	741	722	
	Edh at Noon	88	116	147	169	142	132	134	132	110	99	81	77	
All-Sky Solar Radiation	RadAvg	1.38	2.19	3.12	4.45	5.53	6.28	6.40	5.64	4.32	2.89	1.89	1.28	
	RadStd	0.19	0.28	0.27	0.56	0.60	0.43	0.37	0.36	0.47	0.29	0.23	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+152	N/A
(47)	Regional (5 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-122	+118	N/A

Nomenclature: See separate page